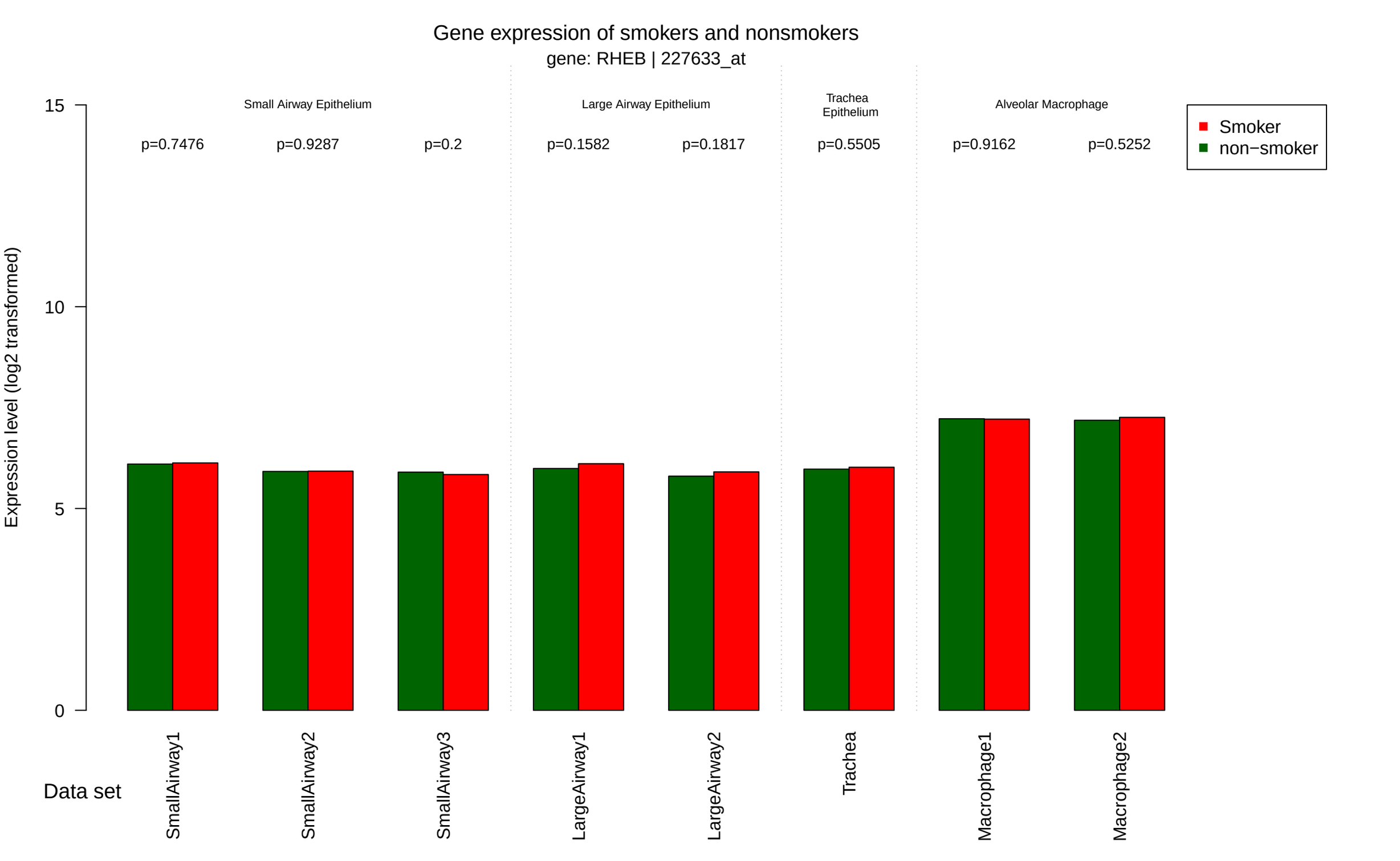
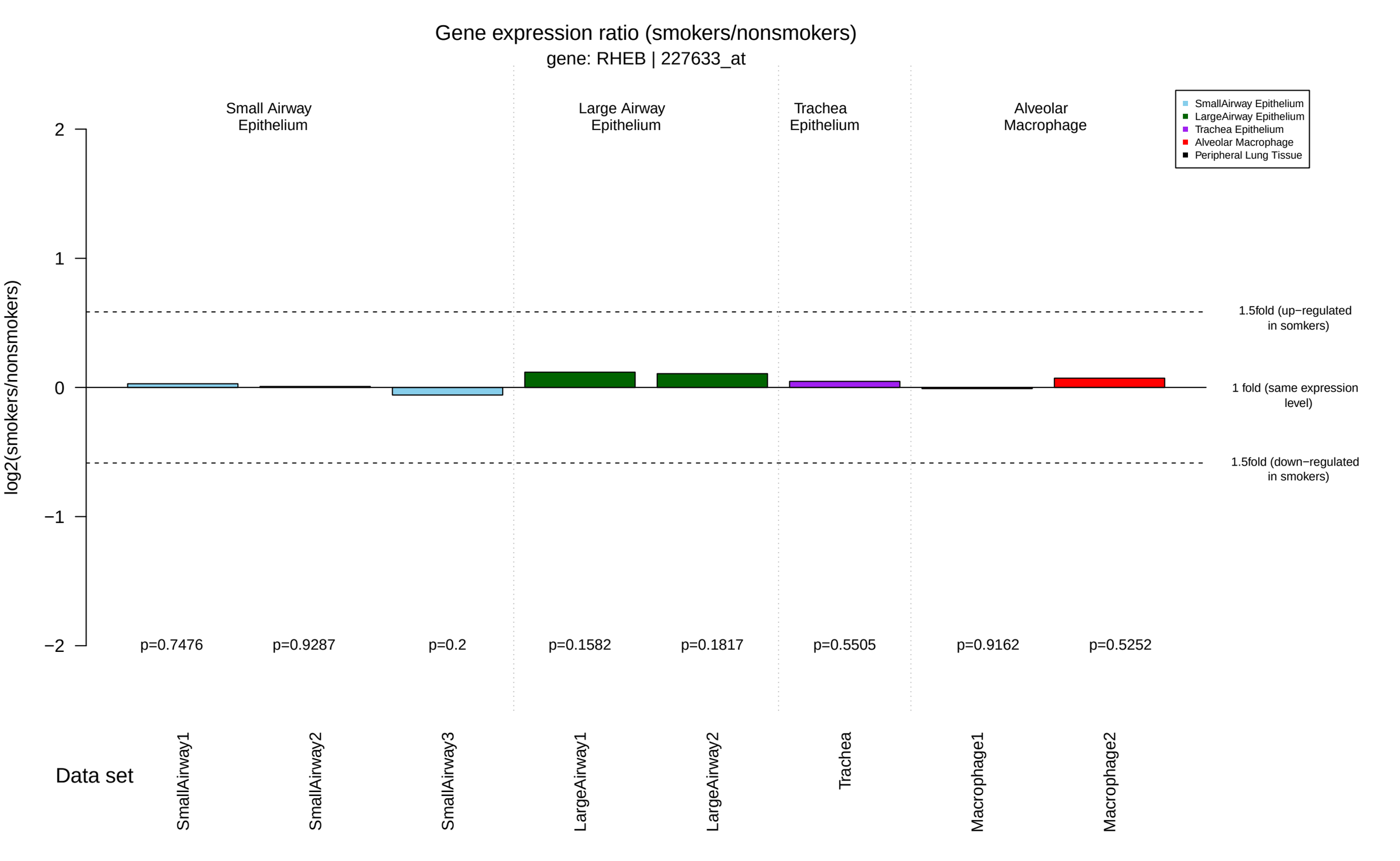
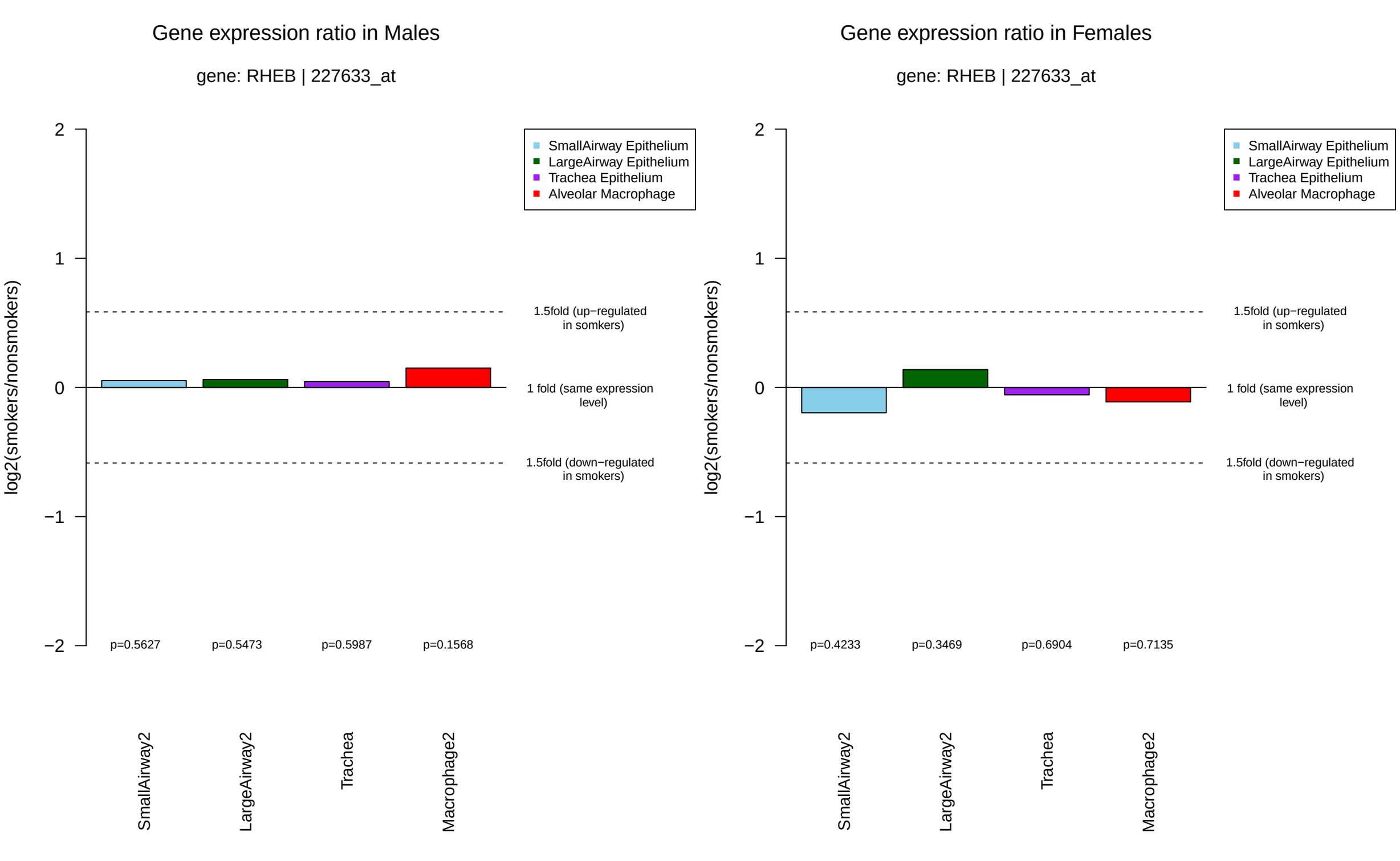
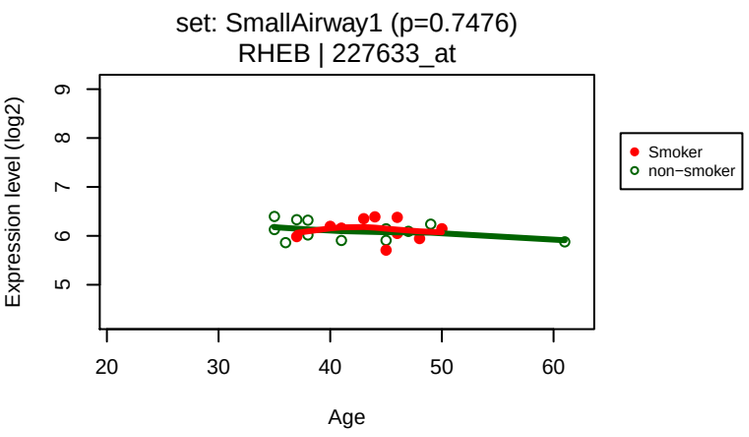
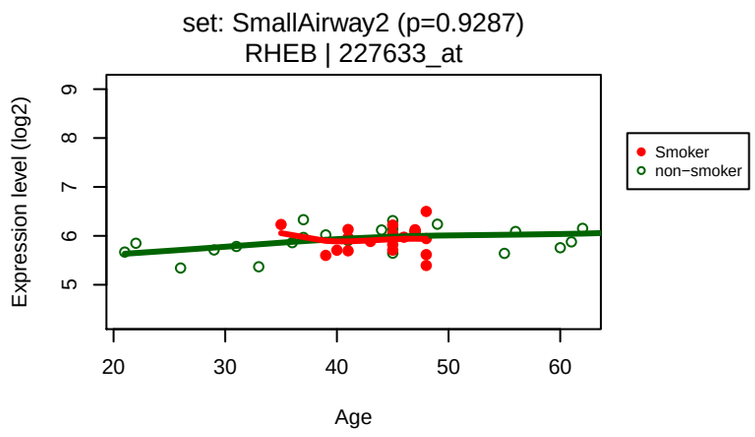
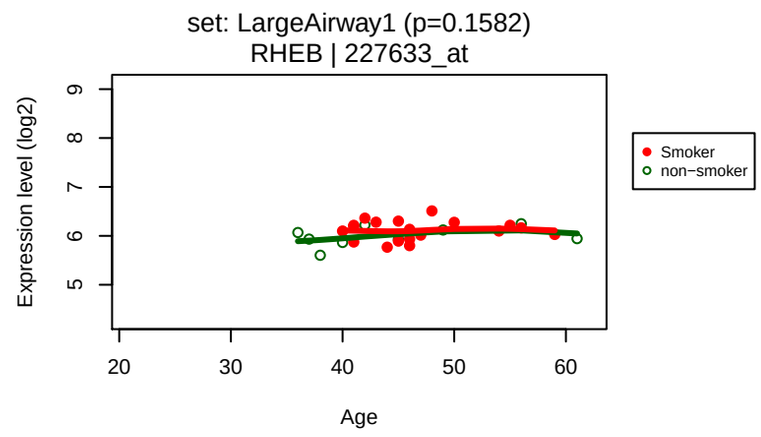

Gene expression of smokers and nonsmokers
gene: RHEB | 227633_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7476
p=0.9287
p=0.2
p=0.1582
p=0.1817
p=0.5505
p=0.9162
p=0.5252
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

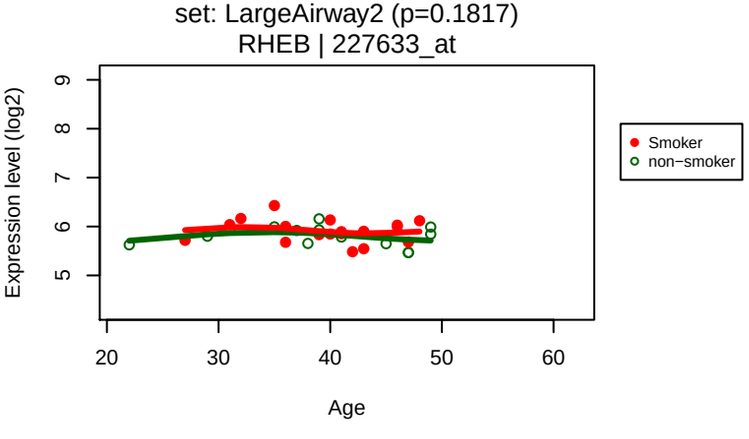

Gene expression ratio (smokers/nonsmokers)
gene: RHEB | 227633_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7476
p=0.9287
p=0.2
p=0.1582
p=0.1817
p=0.5505
p=0.9162
p=0.5252
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

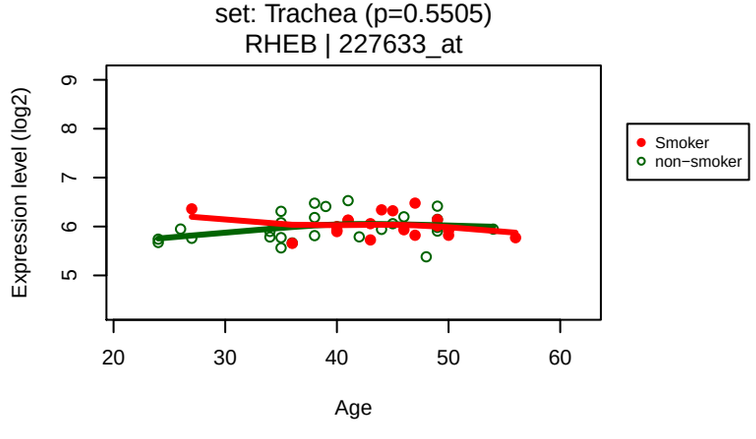

Gene expression ratio in Males
gene: RHEB | 227633_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5627
p=0.5473
p=0.5987
p=0.1568
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHEB | 227633_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4233
p=0.3469
p=0.6904
p=0.7135
SmallAirway2
LargeAirway2
Trachea
Macrophage2

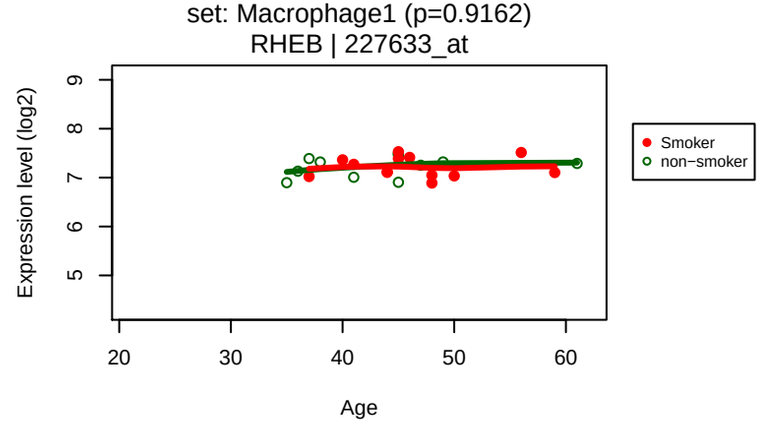

set: SmallAirway1 (p=0.7476)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker

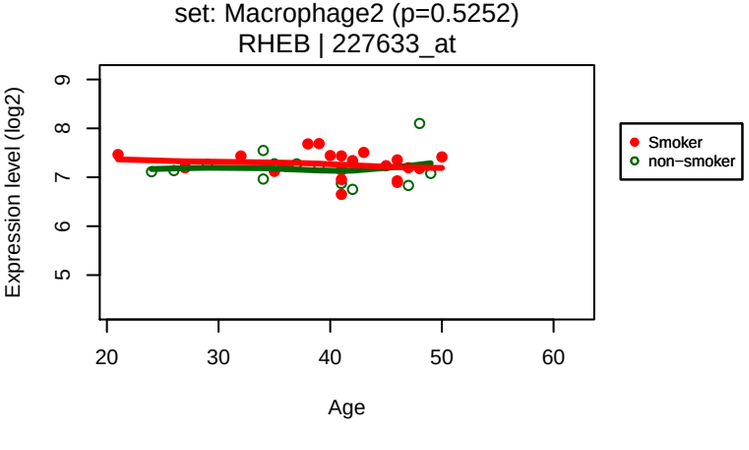

set: SmallAirway2 (p=0.9287)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker

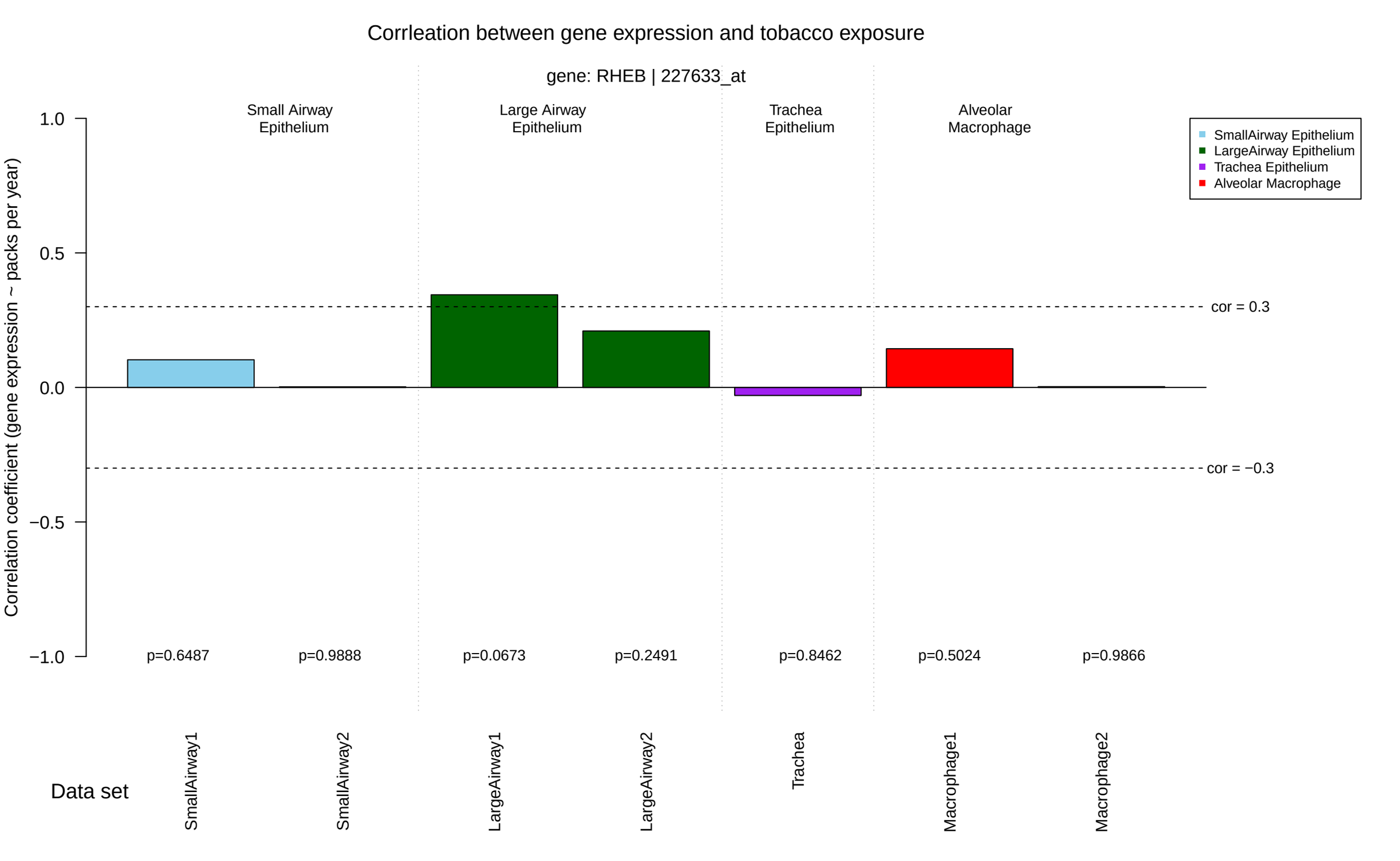

set: LargeAirway1 (p=0.1582)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1817)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5505)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9162)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5252)
RHEB | 227633_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHEB | 227633_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6487
p=0.9888
p=0.0673
p=0.2491
p=0.8462
p=0.5024
p=0.9866
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2